

NHAT MINH PHAM

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EDUCATION

University of Minnesota, College of Science and Engineering Bachelor of Science in Computer Science and Mathematics , Minor in Statistics GPA: 3.9/4.0, Dean's List	Minneapolis, MN <i>Expected May 2028</i>
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TECHNICAL SKILLS

Languages: Python, Go, Java, SQL, C, Bash, JavaScript/TypeScript
Infrastructure & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, ArgoCD, Terraform
Distributed Systems & Data: Kafka, Redis, PostgreSQL, MongoDB
Observability: Prometheus, Grafana, Loki, OpenTelemetry
Systems & Cloud: Linux, QEMU, GDB, GCP (GKE)

RELEVANT EXPERIENCE

VietinBank <i>DevSecOps Engineering Intern</i> - Build Jenkins CI/CD pipelines for 4 Java, Go, Python, and React services on a 9-VM Kubernetes cluster, enforcing security gates with Gitleaks, Semgrep, Trivy, and SonarQube. - Reduce credential exposure by 80% using HashiCorp Vault and harden service-to-service communication with Istio mTLS and zero-trust AuthorizationPolicy. - Implement Kafka/Redis asynchronous event flows and Blue-Green deployments with header-based routing to support zero-downtime releases. - Diagnose Jenkins pipeline failures on VietinBank infrastructure and prototype a local ELK alerting POC that reduces query scan volume by 99% and detects error spikes across 3 services in under 2 minutes.	Hanoi, Vietnam <i>May 2026 - Present</i>
FPT Software <i>Full-Stack Engineering Intern</i> - Built an AI-integrated company research web application using React, TypeScript, Node.js/Express, MongoDB, JWT authentication, REST APIs, Git, and Docker. - Designed and integrated a Retrieval-Augmented Generation pipeline using an LLM API, LangChain, PostgreSQL/pgvector, and sentence-transformers for company-specific semantic search over job postings. - Orchestrated a LangGraph conversational workflow for retrieve → rate → advise and implemented Server-Sent Events to stream AI-generated resume feedback to the React frontend in real time.	Hanoi, Vietnam <i>May 2025 - August 2025</i>

PROJECT EXPERIENCE

MatchSense: Cloud-Native Real-Time Analytics Platform , <i>Independent Project</i> - Architected a distributed platform with 5 Go/Python microservices, Kafka, and Redis, supporting real-time event processing across 3 Kafka topics with prediction updates within 30 ms. - Built a GitOps deployment workflow with ArgoCD managing 12 Kubernetes applications, enforcing 4 Kyverno policies, image digest pinning, and zero cluster credentials in CI. - Implemented end-to-end observability with Prometheus, Loki, and Jaeger, propagating trace IDs through Kafka headers across event-api, event-processor, and ml-predictor services.	June 2026
procmon: Multi-Mechanism Linux Process Monitor , <i>Independent Project</i> - Built three loadable Linux kernel modules in C for system-wide process monitoring using kprobes and stable kernel tracepoints, validating runtime behavior through <code>insmod</code>, <code>dmesg</code>, and kernel symbol inspection. - Developed a Linux misc character-device driver exposing <code>/dev/procmon</code> with blocking <code>read()</code>/<code>poll()</code> and runtime controls, using wait queues and <code>spin_lock_irqsave</code> to synchronize concurrent kernel and userspace operations. - Ported the monitor to eBPF/CO-RE using libbpf and a lock-free BPF ring buffer, resolving tracepoint layout mismatches through BTF inspection with <code>bpftool</code> for portability across kernel versions.	August 2026
Personal Cognitive Load & Focus Monitoring Platform , <i>Independent Project</i> - Provisioned and deployed a containerized FastAPI service on GKE using Terraform, Helm, and KServe, including scale-to-zero serving and infrastructure-as-code workflows. - Built a GitHub Actions CI/CD pipeline with 26 Pytest cases at 92% coverage, container publishing to GHCR, and automated Kubernetes deployment. - Implemented production-style observability with Prometheus metrics, OpenTelemetry tracing, Loki logging, MLflow experiment tracking, and Evidently drift monitoring.	January 2026